



BY SOHAM GAIKWAD
MIT WPU CSE
GMAIL 1262251720@mitwpu.edu

Smart Replacement & Freshness Intelligence System (SRFIS)

A lightweight system to reduce cancellations, improve trust, and increase acceptance of alternatives in grocery fulfillment.

MVP Goal: Enhancing Customer Trust and Efficiency

Reduce Cancellations

Minimize order cancellations caused by customer dissatisfaction with replacements.

Improve Trust

Build customer confidence through transparency in product freshness and replacement logic.

Increase Acceptance

Boost the rate at which customers accept recommended alternative products.

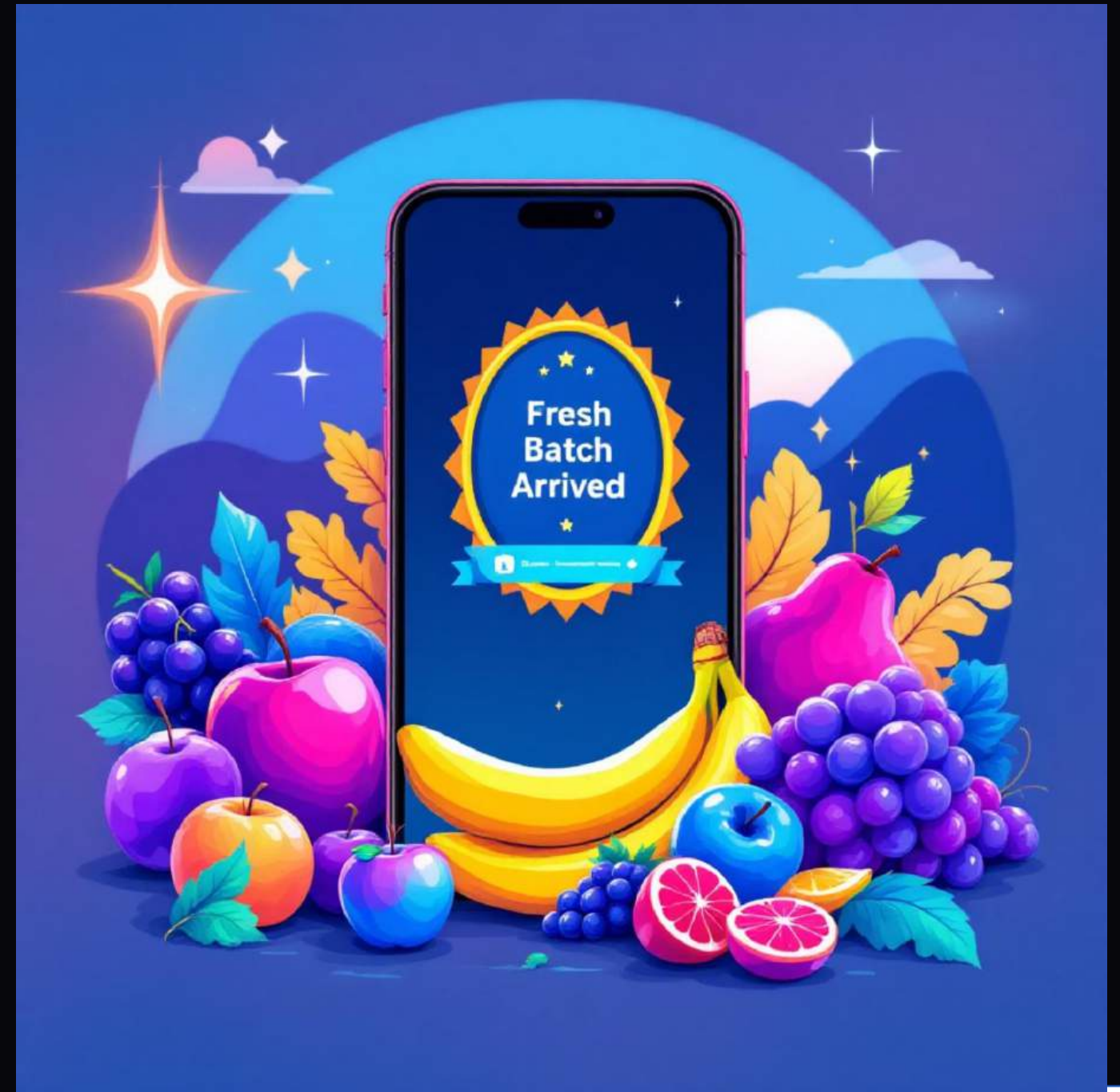


Feature 1: Freshness Indicator

"Fresh Batch Arrived 2 hours ago"

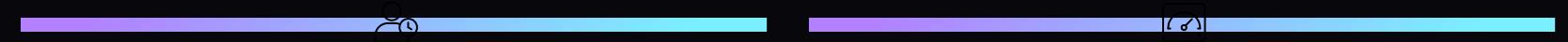
Customers see when highly perishable items (fruits, vegetables, bread, milk) were last restocked.

- Increases confidence in fresh items
- Drives conversion for perishables
- Reduces cancellations due to "quality concerns"



Real-time Batch Arrival Tracking

We're excited to introduce a crucial new feature for Zepto: real-time visibility into batch arrival times. This enhancement addresses a key user need by displaying precisely when newly arrived batches landed, starting with those that arrived two hours ago.



Enhanced Timeliness

Gain immediate insight into recent batch arrivals, improving operational responsiveness.

Improved Oversight

Monitor the flow of batches more effectively, helping to identify and resolve delays swiftly.

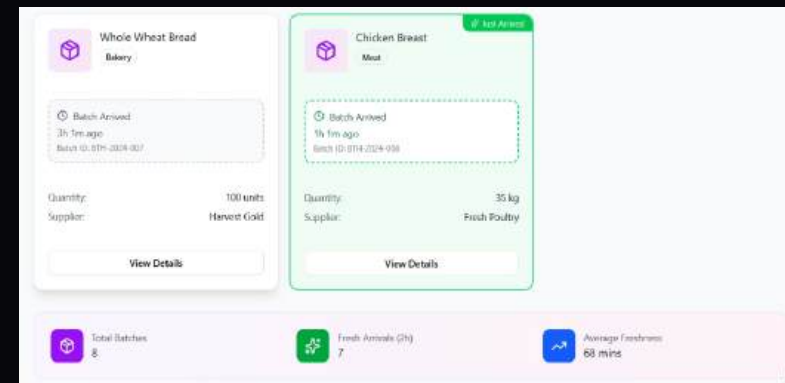
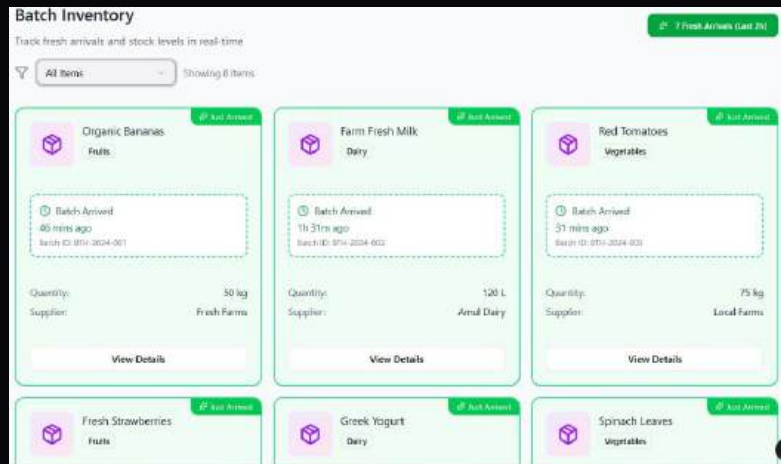


Increased Accuracy

Provide precise timestamps for newly processed batches, ensuring data integrity.

Visualizing the New Feature in Action

This update integrates seamlessly into the existing Zepto interface, offering an intuitive way to track newly arrived batches. The visual representation ensures that critical time data is always front and center, empowering our team to make informed decisions faster.

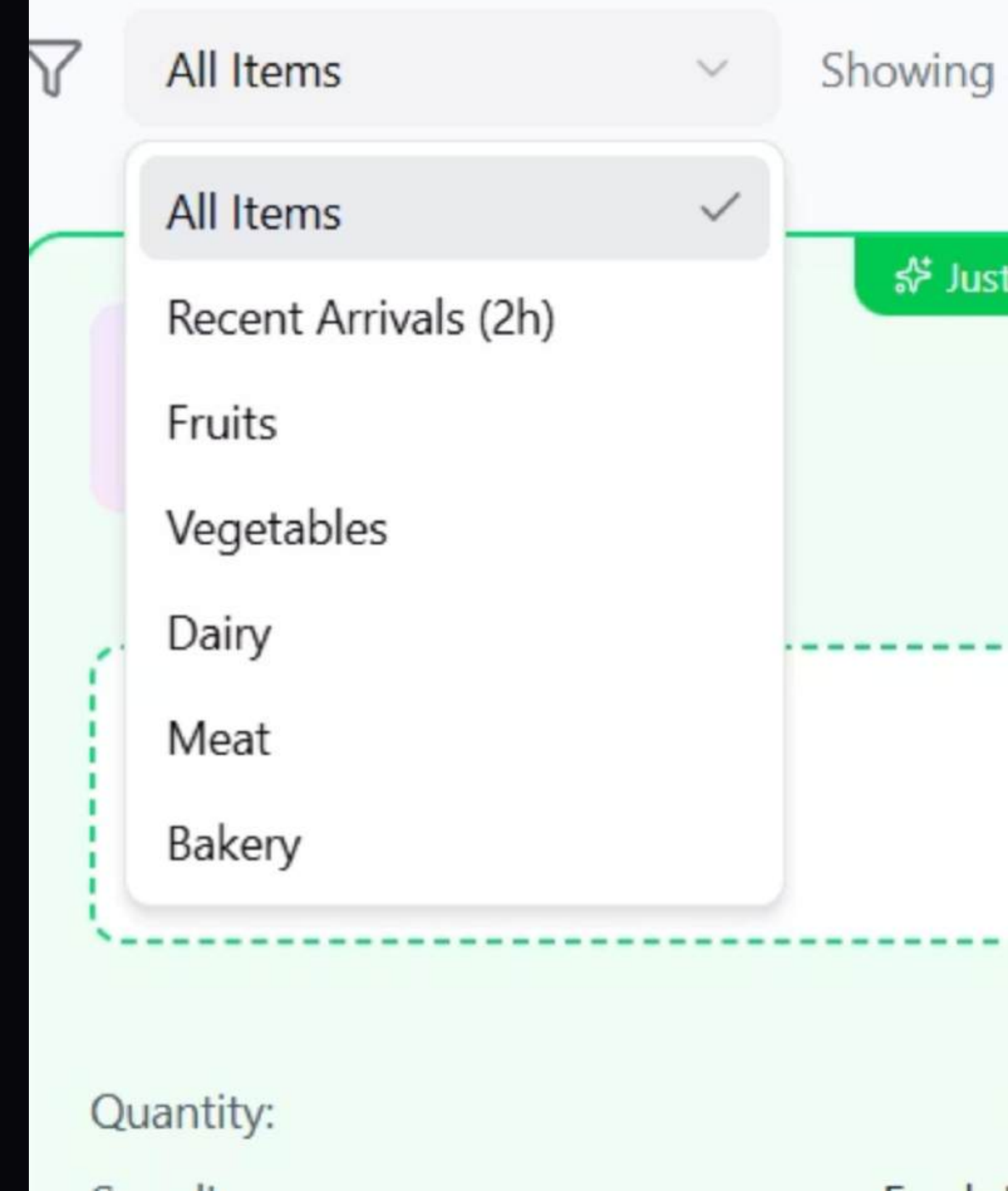


The updated UI highlights batch arrival times prominently, ensuring that the most recent data is easily accessible. This user-centric design aims to streamline workflows and reduce the cognitive load for our users.

This feature provides precise time references for newly processed data batches, improving data transparency and user trust. The subtle visual cues and clear data points minimize potential errors and boost overall system reliability.

Batch Inventory

Track fresh arrivals and stock levels in real-time



Feature 2: Transparent Replacement Reasoning



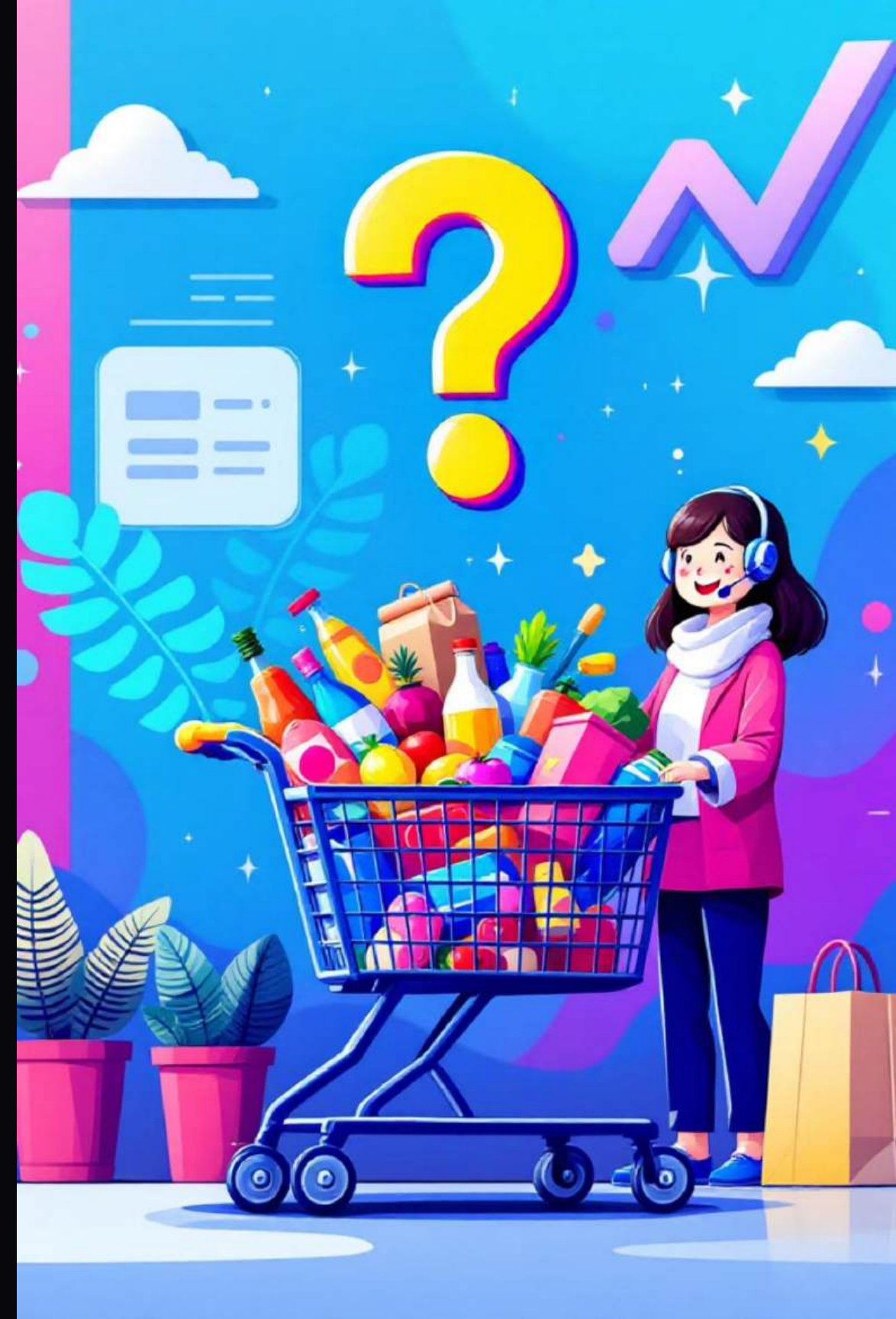
"Why this item is being replaced"

- Original item out of stock at dark store
- Alternative is 99% similar & 2 minutes faster to pick
- Preferred brand not available in selected quantity

This fosters understanding, reduces frustration, and increases replacement acceptance rates.

Navigating Your Order: Understanding Substitutions

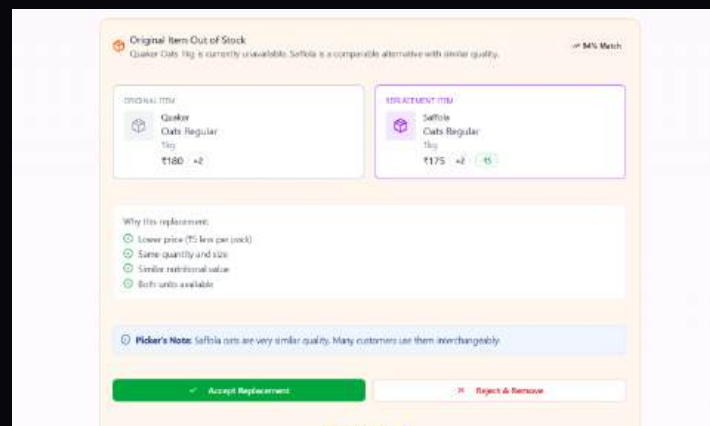
Occasionally, unforeseen circumstances can affect your order. This presentation will explain why substitutions or out-of-stock items occur and what steps we take to ensure your satisfaction.



Why Substitutions Happen

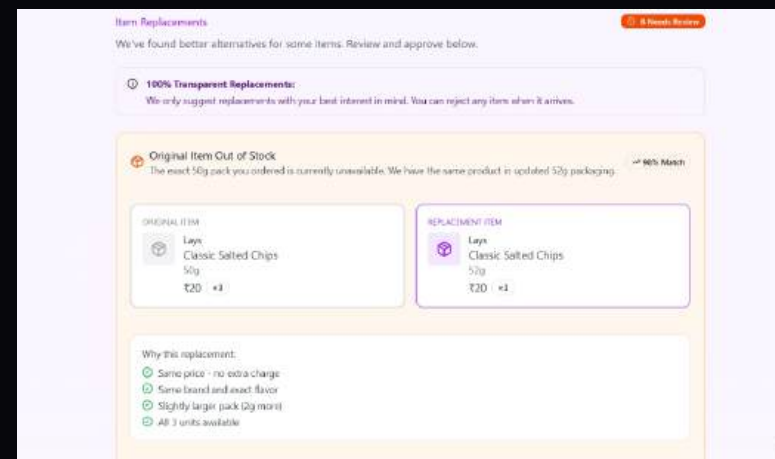
Original Item Out of Stock

The most common reason for a substitution is when the item you originally selected is no longer available. This can be due to high demand, supply chain issues, or unexpected inventory discrepancies.



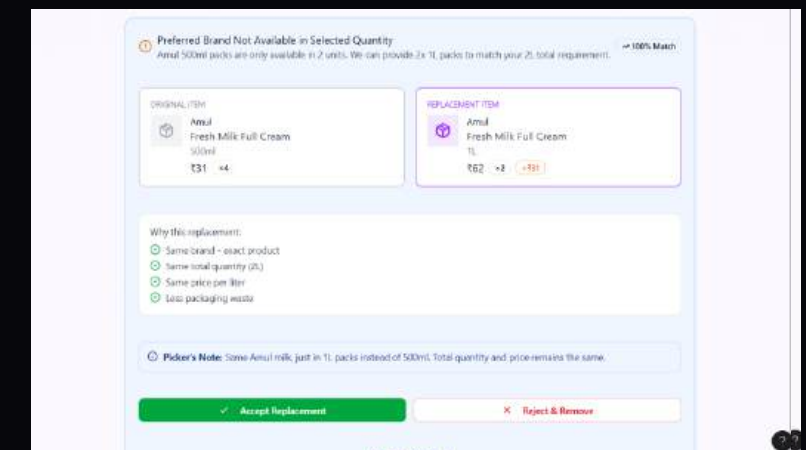
Alternate Item 99% Similar

When a direct match isn't available, we prioritize finding an alternative that is nearly identical to your original choice. This means same brand, same quantity, and similar characteristics to minimize disruption.



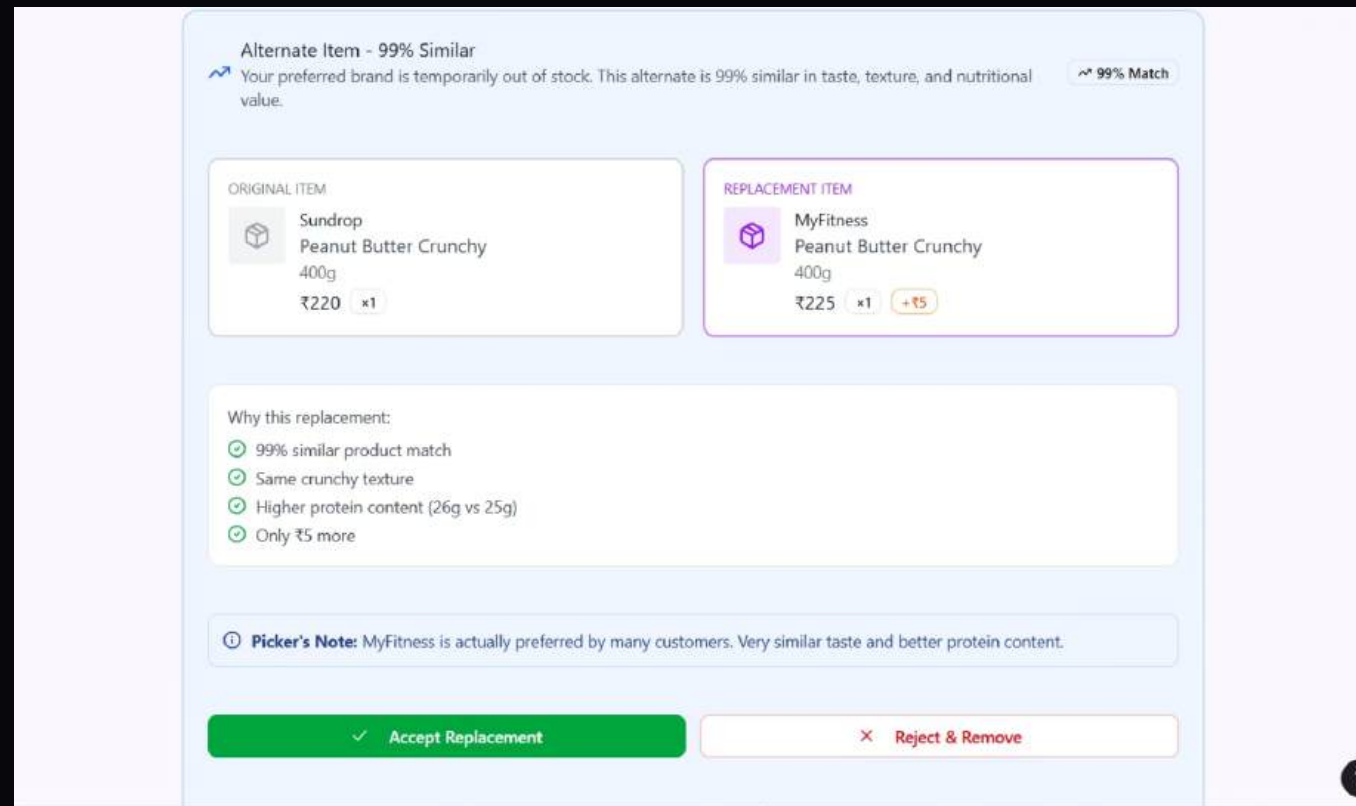
Preferred Brand Unavailable

If your preferred brand is not available in the selected quantity, we'll offer a comparable alternative from a trusted brand to ensure you still receive the necessary product.



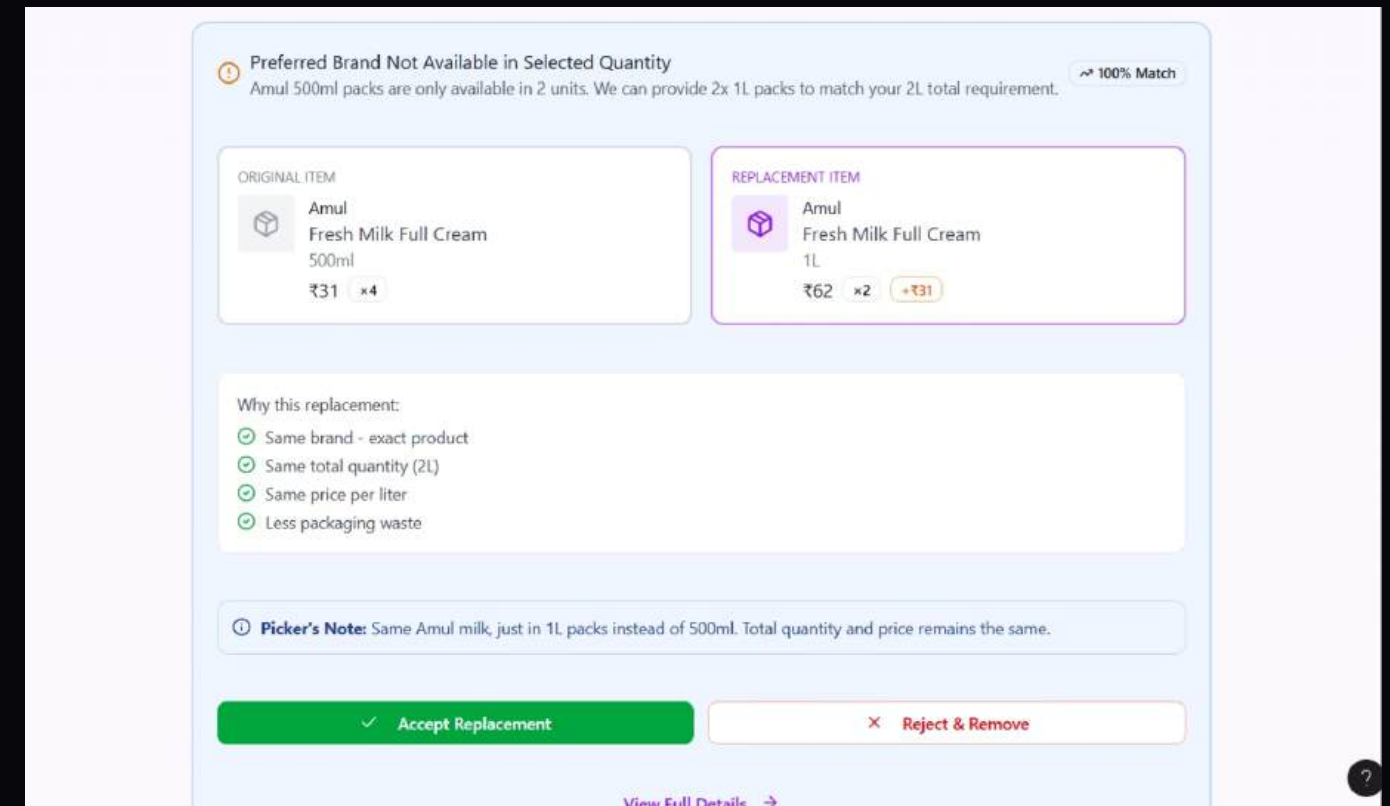
Our Commitment to Your Satisfaction

We understand that substitutions can be inconvenient, but we strive to make the process as seamless as possible. Our goal is to provide the best alternative while keeping you informed every step of the way.



Clear Communication

We'll notify you about any proposed substitutions before your order is finalized, giving you the option to approve or decline.



Quality Alternatives

Our team is trained to select the best possible substitutes, ensuring the quality and integrity of your purchase are maintained.

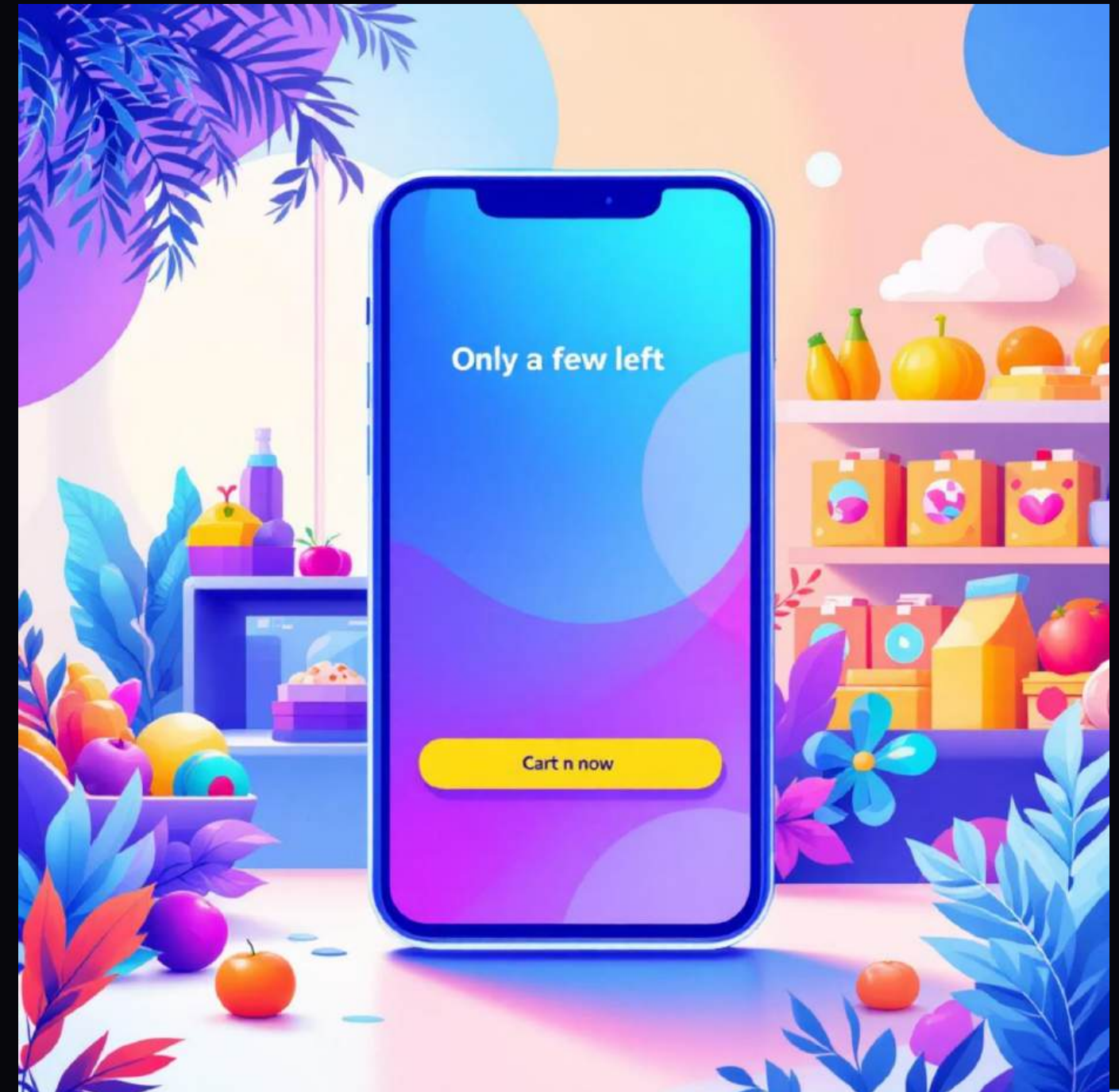
If you ever have concerns about a substitution, our customer service team is ready to assist you. Your satisfaction is our priority!

Feature 3: Predictive Out-of-Stock Alerts

"This SKU may go out of stock soon"

Alerts customers on the cart or product page if an item has a high chance of going out of stock, based on real-time inventory and sell-through rates.

- Reduces surprise replacements
- Pushes users to choose stable SKUs upfront
- Cuts last-minute OOS cancellations



Made with GAMMA

"⚠️ Only a few left — likely to go out of stock soon. 🔄 Consider these alternatives." This

Simplified MVP Architecture



Frontend

Freshness badges, OOS alert banners, and "Why replaced" modals directly engage the customer.



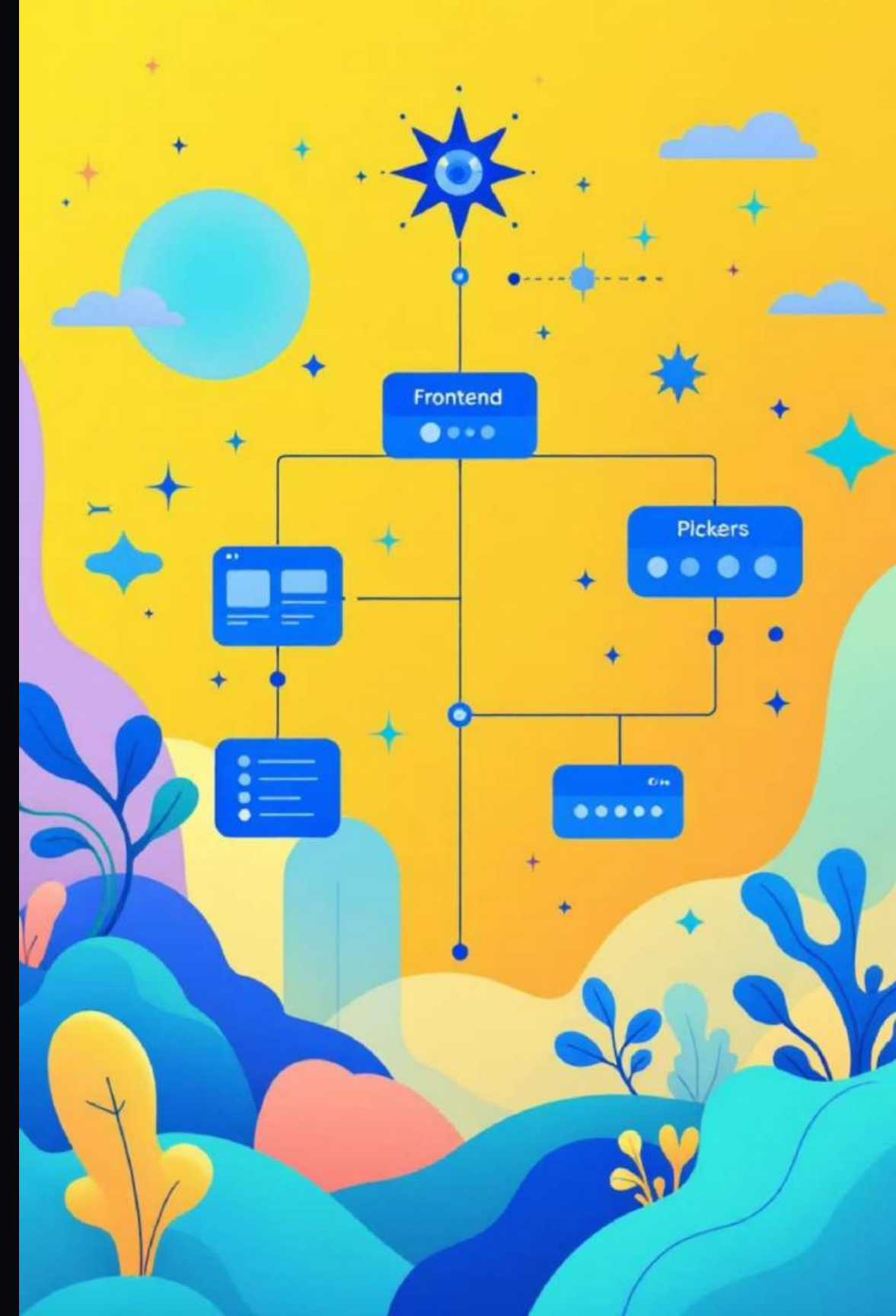
Backend

Manages freshness timestamps, OOS prediction microservices, and replacement reason generation.



Picker App

Logs "Item not found," "Item damaged," or "Out of stock in bin" to inform customer reasons.



MVP Success Metrics: Driving Impact

+12–20% –10–15% +5–7% –8–10%

Replacement
Acceptance Rate

Order Cancellation
Rate

Perishable Item
Conversion

Dark Store OPD
Issues

+5

Customer Trust Score
(CSAT)



Case Study: Problem – High Cancellation Rates

Zepto observed significant order cancellations driven by:

- "I don't trust the replacement. Why was this replaced again?"
- "I wanted fresh stock, not old items."
- "I didn't know it was about to go out of stock."

These issues highlighted a critical need for enhanced transparency and proactive communication.

Case Study: Solution & Impact (4 Weeks Post-MVP)

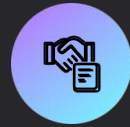
Our hypothesis: Real-time freshness, predictive OOS warnings, and clear replacement reasons would improve acceptance and trust, reducing cancellations.

Replacement acceptance	41%	58%	+17%
Order cancellations	14.5%	11.2%	-3.3%
Perishable category returns	6.2%	4.8%	-1.4%
CSAT	4.1	4.3	+0.2

The **Fresh Batch Indicator** and **Replacement Transparency** were particularly well-received, directly impacting customer satisfaction.



Conclusion: Transparency Fuels Success



Transparency Reduces Anxiety

Clear communication builds confidence and reduces customer frustration.



Freshness Boosts Confidence

Real-time updates on perishables drive conversions and satisfaction.



Predictive Alerts Prevent Issues

Proactive warnings minimize last-minute disruptions and improve user experience.

This MVP successfully validated the problem and demonstrated significant business value with minimal engineering effort, paving the way for future enhancements.